PROJECT WORKFLOW DOCUMENT

NEXTMOVE

From job description to tracked opportunity

Organize your applications.
Track your progress. Move forward.



PROJECT WORKFLOW DOCUMENT

NEXTMOVE

From job description to tracked opportunity

Organize your applications. Track your progress. Move forward.

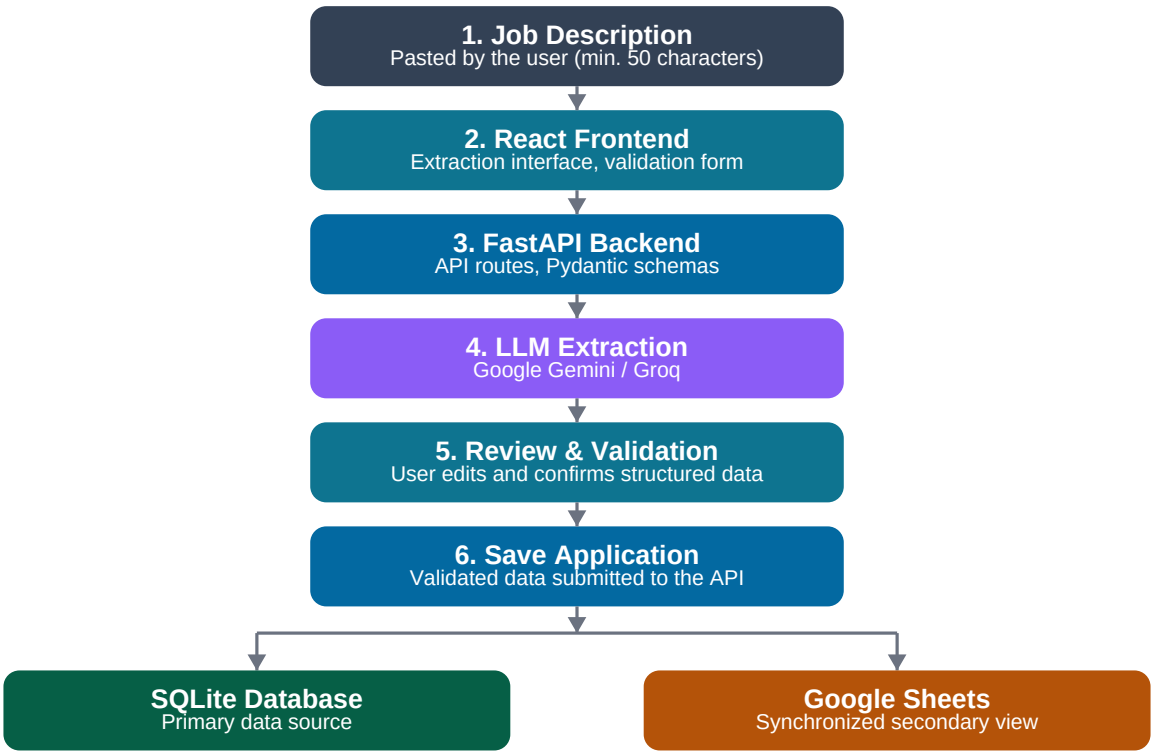
1. Overview

NEXTMOVE is a job application tracking platform. It turns a pasted job description into structured data using AI, lets the user review and correct that data, and then stores the application so its recruitment progress can be tracked. SQLite is the primary data source, and Google Sheets provides an additional synchronized view.

Layer	Technologies
Frontend	React, TypeScript, Vite, Tailwind CSS
Backend	Python, FastAPI, Pydantic
Database	SQLite, SQLAlchemy
AI Integration	Google Gemini, Groq
External Integration	Google Sheets API

2. End-to-End Workflow

The diagram below shows how a job posting travels through the system until it is stored and synchronized.



3. Workflow Steps in Detail

Step 1 - Paste a job description

The user opens the AI Job Extractor page and pastes the full job posting into the text area. The interface shows a live character counter and requires at least 50 characters before extraction is enabled. For the best results the posting should include the company name, job title and location, responsibilities, required qualifications, technical skills and experience requirements.

Step 2 - Send to the backend

The React frontend sends the text to the FastAPI backend through the API integration layer. The backend validates the request with Pydantic schemas before processing it.

Step 3 - LLM extraction

The backend's LLM extraction service sends the text to Google Gemini or Groq and asks for structured output. The model identifies the company, position, location, employment type, required experience, technical and professional skills, and, when available, the job URL and recruitment platform.

Step 4 - Review and validation

The structured result is displayed in editable fields. The user reviews the company, position, location, experience, employment type and required skills, corrects any mistakes, and confirms the details before anything is saved.

Step 5 - Save the application

Once validated, the application is submitted to the backend, which writes it to the SQLite database through SQLAlchemy models. SQLite remains the primary source of truth.

Step 6 - Google Sheets synchronization

After saving, the Google Sheets service, authenticated with a locally stored service account, adds the new application to the configured spreadsheet. Later status updates are propagated to the matching row.

Step 7 - Track progress

In My Applications, the user can search and filter the table, add notes, and update each application's status directly. The dashboard summarizes totals, interviews and offers.

4. Application Statuses

Status	Meaning
Applied	The application has been submitted.
Pending	Waiting for a response or next step.
Interview	The user has been invited to or is in the interview process.
Offer	An offer has been received.
Rejected	The employer declined the application.
Withdrawn	The user withdrew the application.

5. Data Flow and Responsibilities

Component	Responsibility
Job Extraction Interface	Collects the pasted job description and shows extraction progress.
Extraction Validation	Displays editable structured fields so the user can verify data before saving.
Applications Dashboard	Lists applications, supports search and filtering, status updates and key statistics.
API Routes (FastAPI)	Expose endpoints for extraction, saving, listing and updating applications.
Database Models	SQLAlchemy models persisting applications in SQLite.
Pydantic Schemas	Validate incoming and outgoing data.
LLM Extraction Service	Calls Gemini or Groq and returns structured job data.
Google Sheets Service	Appends new applications and updates statuses in the spreadsheet.

6. Running the Project

Backend (from the `backend` folder):

- Create and activate a virtual environment, then install `requirements.txt`.
- Add a `.env` file with `GEMINI_API_KEY`, `GROQ_API_KEY` and `GOOGLE_SHEET_ID`.
- Start the server: `python -m uvicorn app.main:app --reload --port 8000`.
- API docs are available at `http://127.0.0.1:8000/docs`.

Frontend (from the `frontend` folder):

- Run `npm install`, then `npm run dev`, and open the local URL shown.

7. Security Practices

- Keep API keys in local `.env` files and document them in `.env.example`.
- Never commit Google service account credentials.
- Keep local databases and generated files out of version control.

8. Roadmap

Version 1 covers extraction, validation, application management, status tracking, search and Google Sheets sync. Planned next:

- User authentication and personal workspaces
- Advanced analytics dashboard
- Reminders and follow-up notifications
- Automatic job discovery from selected platforms
- AI-powered job description analysis
- CV-to-job matching and skill gap analysis
- Personalized job recommendations
- Production-ready database deployment